

# Tianyi Liu | Dr.-Ing.

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## Profile

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- Current Postdoc and PhD graduate in Electrical Engineering.
- Solid knowledge in signal processing, matrix analysis, and optimization theory.
- Proficiency in parallel/distributed optimization methods in signal processing and machine learning.
- Excellent technical writing skills with 1 book section and 4 journal articles submitted.
- Proven communication and presentation skills fluently in English.

## Education

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### Technical University of Darmstadt

*Dr.-Ing. in Elect. Eng. and Inform. Technol., with distinction*

Advisor: Prof. Marius Pesavento, Co-referee: Prof. Stefan Ulbrich

Committee members: Prof. Abdelhak M. Zoubir, Prof. Sebastian Schöps, and Prof. Mario Kupnik

**Darmstadt, Germany**

*Oct. 2018 – Sept. 2024*

### Technical University of Darmstadt

*M.Sc. in Elect. Eng. and Inform. Technol., with distinction*

Recipient of the Best Master Student Award from the Dept. ETIT.

Master Thesis: *A scalable graph-based mixed-integer linear programming approach for the examination timetabling problem.*

**Darmstadt, Germany**

*Oct. 2016 – July 2018*

### Politecnico di Torino

*M.Sc. in Communications and Computer Networks Eng., with distinction*

**Turin, Italy**

*Sept. 2015 – July 2018*

### Politecnico di Torino

*B.Sc. in Telecommunications Engineering*

**Turin, Italy**

*Sept. 2014 – July 2015*

### Tongji University

*B.Eng. in Electronics and Information Engineering*

**Shanghai, China**

*Sept. 2011 – June 2015*

## Research Interests

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- Sparse signal processing
- Parallel/distributed optimization methods: Focusing on nonconvex and nonsmooth problems
- Sensor array signal processing: Direction-of-arrival estimation
- Graph signal processing: Graph topology inference
- Algorithm unrolling
- Game theory: Generalized Nash equilibrium
- Cooperative localization and control of unmanned systems

## Work Experience

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### Signal Processing Systems Group, TU Delft

*Postdoc Researcher*

Advisor: Prof. Geert Leus

Project: GraSPA (Graph Signal Processing in Action) – graph signal processing in dynamic networks

**Delft, The Netherlands**

*June 2026 – Present*

**Communication Systems Group, TU Darmstadt**

*Guest Researcher*

Job seeking and part-time research.

**Darmstadt, Germany**

*May 2025 – May 2026*

**Communication Systems Group, TU Darmstadt**

*Doctoral and Postdoctoral Research Associate*

**Darmstadt, Germany**

*Sept. 2018 – April 2025*

- PhD Thesis: *A parallel successive convex approximation framework with smoothing majorization for phase retrieval*. Developed a novel parallelizable algorithmic framework based on majorization-minimization (MM) and successive convex approximation (SCA) for a broad class of nonconvex and nonsmooth optimization problems. The main idea is to solve a difficult problem by converting it into a sequence of simpler surrogate problems. It provides the provable convergence to a generalized stationary point. As an application, I employed this algorithmic framework to efficiently address the phase retrieval with dictionary learning problem arising in X-ray diffraction imaging and joint channel estimation and signal detection in MIMO communication systems.
- Collaborative projects on separating signals of different frequencies from their mixtures with sparsity techniques, with application to direction-of-arrival (DOA) estimation using distributed passive sensor arrays.
- Other research works on graphical model learning from real-world data, including meteorological, traffic, financial, and biological data.
- Supervised student projects on analyzing the performance of the GPU implementation with CUDA of the parallel algorithms.
- Programming experiences: Most of the simulations are implemented in MATLAB, C++, or Python, and conducted on Linux-based high-performance computing clusters. Neural networks with PyTorch.
- Co-authored 1 book section, 4 internationally refereed journal articles. Presented my research at 8 international conferences in different countries.

**Politecnico di Torino**

*Research Assistant*

Developed a heuristic algorithm for optimizing constellation in satellite communication. Implemented in C++.

**Turin, Italy**

*Mar. 2015 – July 2015*

## Teaching Experience

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**Communication Systems Group, TU Darmstadt**

*Teaching Assistant*

Prepared and delivered tutorials in fundamental and advanced courses on matrix analysis, optimization theory, MIMO communications, graph signal processing, and machine learning. Supervised 15 individual student projects.

**Darmstadt, Germany**

*Sept. 2018 – April 2025*

## Professional Skills

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**Signal Processing and Communications:** Sparse signal processing/Compressed sensing, Dictionary learning/Sparse coding, Sensor array and statistical signal processing, Source localization/Radar, Direction-of-arrival (DOA) estimation, Beamforming, MIMO communications, Coherent diffraction imaging, Image processing, Graph signal processing, Graph topology learning, Machine learning (Graphical models, Neural networks, Graph neural networks, Algorithm unrolling)

**Optimization and Mathematics:** Parallel and sparse optimization methods, Least-squares and regularized least-squares, Nonlinear programming, Mixed-integer programming (MIP), Matrix analysis, Convex analysis, Graph theory, Probability theory

**Programming:** MATLAB, Python/Numpy/PyTorch, C/C++, Julia, Linux, Bash/Zsh, Git, CUDA

**Optimization Tools:** CVX, MOSEK, Gurobi, SCIP/SCIP-SDP, CPLEX

**Writing and Note Management:** Markdown, LaTeX, Obsidian, Zotero, PARA method, Neovim

**Languages:** Chinese (mother tongue), English (fluent), Italian (intermediate), German (intermediate)

## Honors and Awards

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*Finalist of the IEEE SAM 2024 Best Student Paper Contest*

*2024*

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|----------------------------------------------------------------|------------------|
| <i>IEEE SPS Student Travel Grant for the IEEE CAMSAP 2019</i>  | <i>2019</i>      |
| <i>Finalist of the EUSIPCO 2019 Best Student Paper Contest</i> | <i>2019</i>      |
| <i>Best Master Student Award from Dept. ETIT, TU Darmstadt</i> | <i>2018</i>      |
| <i>Erasmus+ Mobility Program Scholarship</i>                   | <i>2016–2018</i> |
| <i>Full Scholarship from EDISU Piemonte, Italy</i>             | <i>2015–2018</i> |

## Participated Projects

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- German Research Foundation (DFG), “The partial relaxation method in direction-of-arrival estimation: Design and analysis”, PI: Prof. Marius Pesavento, €300K, 2019–2025
- DFG priority program on Compressed Sensing in Information Processing (CoSIP), “Exploiting structure in compressed sensing using side constraints: From analysis to system design – Funding phase II”, PI: Prof. Martin Haardt, Prof. Marc Pfetsch, Prof. Marius Pesavento, €300K for each group, 2018–2021

## Professional Skills Training

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*Attendance at the IEE-SPS/EURASIP Summer School on Network- and Data-Driven Learning* 2019

## Academic Service

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### Reviewer – Journals

- IEEE Transactions on Signal Processing (IEEE TSP)
- IEEE Transactions on Aerospace and Electronic Systems (IEEE TAES)
- IEEE Transactions on Antennas and Propagation (IEEE TAP)
- IEEE Signal Processing Letters (IEEE SPL)
- Elsevier Signal Processing

### Reviewer – Conferences

- IEEE Sensor Array and Multichannel Signal Processing Workshop (IEEE SAM)

## Referees

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**Prof. Marius Pesavento:** Communication Systems Group, TU Darmstadt, Germany  
Email: pesavento@nt.tu-darmstadt.de Phone: +49 (0)6151 16-20342

**Prof. Yonina Eldar:** Department of Electrical and Computer Engineering, Northeastern University, USA  
Email: y.eldar@northeastern.edu

**Prof. Martin Haardt:** Communications Research Laboratory, TU Ilmenau, Germany  
Email: martin.haardt@tu-ilmenau.de Phone: +49 (0)3677 69-2613

**Prof. Abdelhak Zoubir:** Signal Processing Group, TU Darmstadt, Germany  
Email: zoubir@spg.tu-darmstadt.de Phone: +49 (0)6151 16-21341

# Appendix

## Conferences & Workshops

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**Oregon, USA**

*July 2024*

*IEEE 13th Sensor Array and Multichannel Signal Processing Workshop (SAM)*  
Finalist of the Best Student Paper Contest

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| <b>Seoul, Korea, Republic of</b><br><i>IEEE 49th International Conference on Acoustics, Speech, and Signal Processing (ICASSP)</i>                                               | <i>April 2024</i> |
| <b>Herradura, Costa Rica</b><br><i>IEEE 9th International Workshop on Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP)</i>                                    | <i>Dec. 2023</i>  |
| <b>Helsinki, Finland</b><br><i>31st European Signal Processing Conference (EUSIPCO)</i>                                                                                          | <i>Sept. 2023</i> |
| <b>Toronto, Canada</b><br><i>IEEE 46th International Conference on Acoustics, Speech, and Signal Processing (ICASSP)</i>                                                         | <i>June 2021</i>  |
| <b>Hangzhou, China</b><br><i>IEEE 11th Sensor Array and Multichannel Signal Processing Workshop (SAM)</i>                                                                        | <i>June 2020</i>  |
| <b>Guadeloupe, West Indies</b><br><i>IEEE 8th International Workshop on Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP)</i><br>IEEE SPS Student Travel Grant | <i>Dec. 2019</i>  |
| <b>A Coruña, Spain</b><br><i>27th European Signal Processing Conference (EUSIPCO)</i><br>Finalist of the Best Student Paper Contest                                              | <i>Sept. 2019</i> |

## Publications

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### Theses.....

- [T1] **T. Liu**, “A parallel successive convex approximation framework with smoothing majorization for phase retrieval,” Ph.D. dissertation, Technische Universität Darmstadt, Darmstadt, Oct. 2024. DOI: 10.26083/tuprints-00028201
- [T2] **T. Liu**, “A scalable graph-based mixed-integer linear programming approach for the examination timetabling problem,” M.S. thesis, Politecnico di Torino, Turin, Jul. 2018.

### Book Chapters.....

- [B1] K. Ardah, M. Haardt, **T. Liu**, F. Matter, M. Pesavento, and M. E. Pfetsch, “Recovery under side constraints,” in *Compressed sensing in information processing*, G. Kutyniok, H. Rauhut, and R. J. Kunsch, Eds., Cham: Springer International Publishing, 2022, pp. 213–246, ISBN: 978-3-031-09745-4.

### Preprints.....

- [P1] **T. Liu**, S. P. Deram, K. Ardah, M. Haardt, M. E. Pfetsch, and M. Pesavento, *Gridless parameter estimation in partly calibrated rectangular arrays*, Jun. 2024. DOI: 10.48550/arXiv.2406.16041 arXiv: 2406.16041 [eess].

### Journal Articles.....

- [J1] **T. Liu**, F. Matter, A. Sorg, M. E. Pfetsch, M. Haardt, and M. Pesavento, “Maximum a posteriori direction-of-arrival estimation via mixed-integer semidefinite programming,” *IEEE Transactions on Signal Processing*, vol. 74, pp. 2503–2518, 2026. DOI: 10.1109/TSP.2026.3697636
- [J2] R. Müller, G. Allevato, M. Rutsch, C. Haugwitz, **T. Liu**, M. Kupnik, and M. Pesavento, “A tensor model for the calibration of air-coupled ultrasonic sensor arrays in 3D imaging,” *Signal Processing*, p. 109812, Nov. 2024, ISSN: 0165-1684. DOI: 10.1016/j.sigpro.2024.109812
- [J3] **T. Liu**, A. M. Tillmann, Y. Yang, Y. C. Eldar, and M. Pesavento, “Extended successive convex approximation for phase retrieval with dictionary learning,” *IEEE Transactions on Signal Processing*, vol. 70, pp. 6300–6315, 2022, ISSN: 1941-0476. DOI: 10.1109/TSP.2022.3233253

## Conference Proceedings.....

- [C1] **T. Liu**, S. P. Deram, K. Ardah, M. Haardt, M. E. Pfetsch, and M. Pesavento, "Gridless parameter estimation in partly calibrated rectangular arrays," in *ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, Apr. 2024, pp. 8796–8800. DOI: 10.1109/ICASSP48485.2024.10446959
- [C2] **T. Liu** and M. Pesavento, "Blind phase-offset estimation in sparse partly calibrated arrays," in *2024 IEEE 13rd Sensor Array and Multichannel Signal Processing Workshop (SAM)*, Jul. 2024, pp. 1–5. DOI: 10.1109/SAM60225.2024.10636507
- [C3] **T. Liu**, F. Matter, A. Sorg, M. E. Pfetsch, M. Haardt, and M. Pesavento, "Joint sparse estimation with cardinality constraint via mixed-integer semidefinite programming," in *2023 IEEE 9th International Workshop on Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP)*, Herradura, Costa Rica, Dec. 2023, pp. 106–110. DOI: 10.1109/CAMSAP58249.2023.10403415
- [C4] Y. Zhang, **T. Liu**, and M. Pesavento, "Direction-of-arrival estimation for correlated sources and low sample size," in *2023 31st European Signal Processing Conference (EUSIPCO)*, Sep. 2023, pp. 1559–1563. DOI: 10.23919/EUSIPCO58844.2023.10290019
- [C5] **T. Liu**, A. M. Tillmann, Y. Yang, Y. C. Eldar, and M. Pesavento, "A parallel algorithm for phase retrieval with dictionary learning," in *IEEE International Conference on Acoustics, Speech and Signal Processing*, Jun. 2021, pp. 5619–5623. DOI: 10.1109/ICASSP39728.2021.9413991
- [C6] X. Wang, **T. Liu**, M. Trinh-Hoang, and M. Pesavento, "GPU-accelerated parallel optimization for sparse regularization," in *2020 IEEE 11th Sensor Array and Multichannel Signal Processing Workshop (SAM)*, Jun. 2020, pp. 1–5. DOI: 10.1109/SAM48682.2020.9104328
- [C7] **T. Liu**, M. Trinh-Hoang, Y. Yang, and M. Pesavento, "A block coordinate descent algorithm for sparse Gaussian graphical model inference with laplacian constraints," in *IEEE International Workshop on Computational Advances in Multi-Sensor Adaptive Processing*, Dec. 2019, pp. 236–240. DOI: 10.1109/CAMSAP45676.2019.9022643
- [C8] **T. Liu**, M. Trinh-Hoang, Y. Yang, and M. Pesavento, "A parallel optimization approach on the infinity norm minimization problem," in *2019 27th European Signal Processing Conference (EUSIPCO)*, A Coruna, Spain: IEEE, Sep. 2019, pp. 1–5, ISBN: 978-90-827970-3-9. DOI: 10.23919/EUSIPCO.2019.8902548

## Teaching Assistant at TU Darmstadt

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### Courses.....

#### 18-pe-2080 Graph Signal Processing, Learning and Optimization

Instructor: Prof. Marius Pesavento

WS 2020/21 – WS 2024/25

#### 18-pe-2070 Matrix Analysis and Computations

Instructor: Prof. Marius Pesavento

SS 2019 – SS 2024

#### 18-pe-2050 Project Seminar: Procedures for Massive MIMO and 5G

Instructor: Prof. Marius Pesavento

SS 2020

#### 18-pe-2030 MIMO Communication and Space-Time Coding

Instructor: Prof. Marius Pesavento

WS 2018/19 – WS 2019/20

### Supervised Student Projects.....

- [S1] Y. Hou, "Gridless sparse DoA estimation with shift invariant subarrays using stochastic maximum likelihood," M.S. thesis, Nov. 2025.
- [S2] A. Ahmad, "Distributed nonlinear polynomial graph filter and its output graph spectrum," Project seminar, Technische Universität Darmstadt, Darmstadt, Dec. 2022.
- [S3] A. Sorg, "Direction of arrival estimation in the multiple measurement model using sparse reconstruction," M.S. thesis, Technische Universität Darmstadt, Darmstadt, Nov. 2022.

- [S4] Y. Zhang, “Advanced optimization methods for direction-of-arrival estimation with correlated sources,” M.S. thesis, Technische Universität Darmstadt, Darmstadt, Mar. 2022.
- [S5] S. Rajurkar, “Resource allocation in wireless networks using graph convolutional neural networks,” M.S. thesis, Technische Universität Darmstadt, Darmstadt, Nov. 2021.
- [S6] H. S. Saka, “Sparse graph Laplacian learning with real-world data,” Bachelor thesis, Technische Universität Darmstadt, Darmstadt, Apr. 2021.
- [S7] Y. Zhang, “Sparse graph Laplacian learning with real-world data,” M.S. thesis, Technische Universität Darmstadt, Darmstadt, May 2021.
- [S8] S. Ben Abid, “Parallel optimization methods for graph topology learning and tracking,” Bachelor thesis, Technische Universität Darmstadt, Darmstadt, Mar. 2020.
- [S9] F. Bonakdar, “Sparse reconstruction and prediction of mobility patterns in traffic networks,” M.S. thesis, Technische Universität Darmstadt, Darmstadt, Jun. 2020.
- [S10] M. Grytz, “Implementation of a block coordinate descent graph learning algorithm and its application to real-world problems,” Studienarbeit, Technische Universität Darmstadt, Darmstadt, Oct. 2020.
- [S11] H. S. Saka, “Optimization algorithms for graph Laplacian estimation via ADMM and MM,” Project seminar, Technische Universität Darmstadt, Darmstadt, Sep. 2020.
- [S12] D. Jaoua, “Parallel optimization methods for sparse signal recovery from non linear mixtures,” M.S. thesis, Technische Universität Darmstadt, Darmstadt, 2019.
- [S13] S. U. Rehman, “Implementation of a parallel algorithm for sparse optimization on a graphical processing unit,” M.S. thesis, Technische Universität Darmstadt, Darmstadt, Jul. 2019.
- [S14] X. Wang, “Implementation of parallel optimization for nondifferentiable nonconvex problems on GPU architecture,” M.S. thesis, Technische Universität Darmstadt, Darmstadt, Sep. 2019.